

PAPER First-Principles Near-Neutral Low-Activation Refractory Alloys for Fusion Plasma-Facing Components: Formation-Energy Screening and Entropy-Stabilized Composition-of-Matter Design

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Official version: <https://www.kronosfusionenergy.com/publications/low-activation-alloys.html>

Abstract

Structural and plasma-facing alloys that activate little and clear their radiological classification within a human lifetime are a hard requirement for a compact fusion breeder and the natural home of composition-of-matter design. We report a first-principles screen of refractory multi-principal-element alloys for the first-wall, divertor, breeder-blanket, and direct-energy-converter (DEC) electrode roles of the Hyperion compact spherical-tokamak breeder (design point $Q = 3.076$, $P_{\text{fus}} = 85.04$ MW, $I_p = 9.66$ MA, negative triangularity $\delta = -0.30$, $B_0 = 8$ T, elongation $\kappa = 2.0$, $R_0 = 1.20$ m, aspect ratio $A = 2.5$, centrepost peak field 16.84 T), where the first wall accumulates 18.85 dpa per full-power year. A machine-learned interatomic potential (CHGNet) screens a 7500-composition space and returns eight leads that we confirm by plane-wave density-functional theory (DFT; Quantum ESPRESSO, PBE) on special-quasirandom-structure (SQS) cells. The confirmed lead compositions are *near-neutral* in formation energy relative to their body-centred-cubic elemental references: the divertor alloy $\text{Cr}_1\text{Ta}_5\text{Ti}_3\text{V}_3\text{W}_4$ at +0.015 eV/atom (two independent codes agree, +0.0147 and +0.016), the first-wall alloy $\text{Cr}_1\text{Ta}_4\text{Ti}_5\text{V}_3\text{W}_3$ at +0.007, an equiatomic W–Ta–V–Ti refractory high-entropy candidate at +0.003, and a blanket alloy at +0.021 eV/atom; a tantalum-free sub-family chosen to remove the long-lived activation tail spans -0.002 to -0.015 eV/atom. We show that once the ideal configurational mixing entropy is included, every lead has a negative Gibbs mixing free energy ($\Delta G_{\text{mix}} \approx -0.11$ to -0.12 eV/atom at 1000 K)—these are entropy-stabilized single-BCC candidates, with valence-electron concentrations of 4.88–5.31 deep inside the BCC solid-solution window—while being enthalpically metastable-to-marginal rather than deeply stable single phases. We therefore state the invention basis honestly: it is the entropy-stabilized design-and-processing method that realizes and retains these low-activation compositions, not an intrinsic-enthalpy-stability claim. The screen uses a coarse SQS cell; a dense-cell confirmation is carried openly as a second wave, and no composition is presented here as a validated stable phase.

Keywords: low-activation alloys; refractory high-entropy alloys; density-functional theory; formation energy; special quasirandom structures; machine-learned interatomic potentials; fusion plasma-facing components

1 Introduction

1.1 Why low-activation refractory alloys, and why now

A compact fusion breeder that runs on a deuterium–tritium plasma exposes its plasma-facing components (PFCs) and first-wall structure to a 14.1 MeV neutron spectrum. Two consequences dominate materials selection. First, the neutrons displace lattice atoms; at the Hyperion first wall the displacement rate is 18.85 dpa per full-power year, and each displacement cascade seeds the microstructural evolution—voids, bubbles, dislocation loops, and precipitation—that limits component life [1, 2]. Second, transmutation and neutron capture convert the constituent elements into radioactive daughters, some of which are long-lived; the resulting inventory sets whether a decommissioned component can be cleared or must be managed as long-lived waste on a geological timescale [3, 4]. A fusion system whose waste is to clear within a human lifetime must therefore be built, wherever the neutron flux is significant, out of *low-activation* elements combined so that no long-lived activation product accumulates [5, 2].

The classical low-activation palette for fusion structure is narrow: reduced-activation ferritic–martensitic (RAFM) steels [6], vanadium alloys [7], silicon carbide composites, and, for the highest-flux surfaces, tungsten and its alloys [8]. Each element must be chosen not only for its mechanical and thermal performance but for the half-lives of its transmutation products: niobium (producing ^{94}Nb), molybdenum, nickel, cobalt, and—for the divertor—rhenium, are the elements whose activation tails frustrate clearance. The design problem is combinatorial: the requirement is a single-phase, mechanically robust, refractory solid solution whose constituent set is restricted to activation-benign elements. This is precisely the territory of refractory high-entropy alloys (HEAs) [9, 10, 11]: multi-principal-element body-centred-cubic (BCC) solid solutions stabilized in part by configurational entropy, several of which have shown outstanding radiation tolerance [12] and have been fabricated for fusion PFC use [13].

1.2 Prior art: from Hume-Rothery rules to first-principles and machine-learned screening

Phase selection in multicomponent alloys has historically been guided by empirical parameters: the valence-electron concentration (VEC) that discriminates BCC from FCC solid solutions [16], the atomic-size mismatch δ_r and mixing enthalpy ΔH_{mix} that bound single-phase formation [17], and the entropy–enthalpy competition that names the HEA class [14, 15]. These rules are fast but coarse. The quantitative ground truth for phase stability is the density-functional-theory (DFT) formation energy [18, 19], which for a disordered alloy is most economically evaluated on a special-quasirandom structure (SQS) [24, 25] that reproduces the pair- and multi-site correlation functions of the random solid solution in a small periodic cell. DFT, however, is expensive, so the modern practice is to screen a very large composition space with a fast surrogate—a machine-learned interatomic potential (MLIP) [26, 27]—and confirm only the leads at the DFT level. This MLIP-screen/DFT-confirm strategy is now standard in computational alloy discovery and is the backbone of the present work.

1.3 Contribution

We apply an MLIP-screened, DFT-confirmed, activation-filtered pipeline to the specific structural roles of the Hyperion breeder and its companion tandem-mirror burner’s direct-energy converter. Our contributions are:

1. A formal statement of the coupled screening problem—electronic-structure ground state, SQS ensemble average, mixing free energy, and the activation and displacement constraints—as a set of governing equations (§2).
2. A GPU-accelerated computational campaign (§3) that screens 7500 candidate compositions by MLIP and confirms eight leads by plane-wave DFT on SQS cells, with two independent plane-wave codes agreeing on the divertor lead.
3. The central quantitative result (§4): the confirmed leads are *near-neutral* in formation energy (-0.015 to $+0.021$ eV/atom), a result that refutes an earlier spurious -0.172 eV/atom screen value and re-bases the intellectual property from intrinsic enthalpy stability onto the entropy-stabilized processing method.
4. A demonstration that the ideal configurational mixing entropy drives every lead to a negative Gibbs mixing free energy at service temperature, with valence-electron concentrations placing all leads inside the single-BCC window—the physical basis for calling these entropy-stabilized single-phase candidates.

The design point, the displacement rate, and the activation waste-class behaviour are frozen anchors of the Kronos physics de-risking programme; the breeder platform that defines them is described in the companion platform paper [33], its PFC lifetime under 14 MeV loading in [34], its component activation and waste-class routing in [35], and the plasma-material erosion budget in [36]. The MLIP-to-activation pipeline is developed at length in [37], the radiation-tolerant HEA processing route in [38], the tantalum- and rhenium-free family in [39], and the correlated-electron chemistry beyond DFT in [40].

2 Governing equations and the formal screening problem

2.1 Electronic-structure ground state

The primitive quantity is the total energy of a candidate supercell in its electronic ground state. Within Kohn–Sham DFT [18, 19] the interacting many-electron problem is mapped onto

non-interacting particles moving in an effective potential:

$$\left[-\frac{\hbar^2}{2m} \nabla^2 + V_{\text{ext}}(\mathbf{r}) + V_H[n](\mathbf{r}) + V_{xc}[n](\mathbf{r}) \right] \phi_i(\mathbf{r}) = \varepsilon_i \phi_i(\mathbf{r}), \quad (1)$$

with the electron density and the Hartree potential closing the self-consistency loop,

$$n(\mathbf{r}) = \sum_i^{\text{occ}} |\phi_i(\mathbf{r})|^2, \quad V_H[n](\mathbf{r}) = e^2 \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3r', \quad (2)$$

and the exchange–correlation potential $V_{xc} = \delta E_{xc}[n]/\delta n$ taken in the Perdew–Burke–Ernzerhof (PBE) generalized-gradient approximation [20]. The total energy $E[n]$ is evaluated on a plane-wave basis with pseudopotentials; Brillouin-zone integrals use a Monkhorst–Pack mesh [21], and the atomic positions and cell shape are relaxed by variable-cell BFGS to the force/stress ground state.

2.2 Formation energy and the SQS ensemble average

Phase stability of a candidate composition is screened by the formation energy per atom relative to the elemental references in their equilibrium (here BCC) structures,

$$E_f = \frac{1}{N} \left(E_{\text{alloy}} - \sum_i x_i E_i^{\text{bulk}} \right), \quad \sum_i x_i = 1, \quad (3)$$

where N is the number of atoms in the cell, x_i the mole fraction of species i , and E_i^{bulk} the per-atom energy of the elemental reference computed at the same level of theory (e.g. the BCC-W reference -12.7625 eV/atom used as a smoke test; the spin-polarized BCC-Cr reference -248.219 Ry per cell). A negative E_f indicates the mixture is enthalpically favoured against phase separation into the elements; a value within a thermal scale $k_B T$ of zero is *near-neutral*, i.e. metastable-to-marginal.

Because the target is a chemically disordered solid solution, E_{alloy} is the configurational average over site decorations. We evaluate it on a special quasirandom structure [24, 25], the single periodic cell whose truncated set of correlation functions $\{\bar{\Pi}_\alpha^{\text{SQS}}\}$ best matches those of the ideal random alloy $\{\langle \Pi_\alpha \rangle_{\text{rand}}\}$:

$$\text{SQS} = \arg \min_{\sigma} \sum_{\alpha} w_{\alpha} |\Pi_{\alpha}(\sigma) - \langle \Pi_{\alpha} \rangle_{\text{rand}}|, \quad \langle \Pi_{\alpha} \rangle_{\text{rand}} = \prod_{k \in \alpha} (2x_k - 1), \quad (4)$$

with σ the site occupation, α ranging over pair and multi-site clusters, and w_{α} a decreasing weight with cluster size. Equation (4) replaces an average over exponentially many decorations by a single small-cell calculation; the residual finite-size error in E_f is the principal reason a coarse cell must be confirmed by a denser one (§6).

2.3 Entropy stabilization and the mixing free energy

Formation energy is an enthalpy proxy at 0 K; phase stability at service temperature is governed by the Gibbs free energy of mixing. For an ideal (regular-solution, random) multi-principal-element alloy the configurational entropy is

$$\Delta S_{\text{mix}} = -R \sum_i x_i \ln x_i, \quad (5)$$

so that the mixing free energy per mole of atoms, neglecting vibrational and short-range-order corrections, is

$$\Delta G_{\text{mix}}(T) = \Delta H_{\text{mix}} - T \Delta S_{\text{mix}} \approx E_f \mathcal{N}_A - T \Delta S_{\text{mix}}, \quad (6)$$

identifying the DFT formation energy with the mixing enthalpy proxy. Equation (6) makes the central design statement quantitative: a composition that is only *near-neutral* in E_f can still be free-energy-stabilized as a single solid solution provided $T \Delta S_{\text{mix}}$ exceeds ΔH_{mix} and no competing ordered compound lies below ΔG_{mix} . For an equiatomic n -component alloy $\Delta S_{\text{mix}} = R \ln n$; for $n = 4$ this is $1.386R = 11.53 \text{ J mol}^{-1} \text{ K}^{-1}$, worth $-T \Delta S_{\text{mix}} = -0.120$ eV/atom at 1000 K, which dwarfs the $+0.003$ to $+0.021$ eV/atom formation energies of the leads. This is the physics behind “entropy-stabilized” in the title.

2.4 Solid-solution phase-selection constraints

Two empirical constraints filter the composition space before and after the DFT confirmation. The valence-electron concentration,

$$\text{VEC} = \sum_i x_i (\text{VEC})_i, \quad (7)$$

discriminates the crystal structure of the solid solution: $\text{VEC} < 6.87$ favours a single BCC phase, with FCC increasingly favoured above [16]. The atomic-size mismatch and mixing-enthalpy window ($\delta_r, \Delta H_{\text{mix}}$) bound single-phase versus intermetallic formation [17]. We use equation (7) as a post-hoc consistency check on the DFT leads (§4).

2.5 The neutron-irradiation and activation constraints

The alloy must survive and clear a 14 MeV neutron environment. The displacement rate follows the Norgett–Robinson–Torrens (NRT) standard [29],

$$R_{\text{dpa}} = \frac{1}{N} \int \sigma_d(E) \phi(E) dE, \quad \sigma_d(E) = \frac{0.8 T_{\text{dam}}(E)}{2E_d} \sigma_{\text{el}}(E), \quad (8)$$

with $\phi(E)$ the neutron flux spectrum, E_d the displacement threshold, and T_{dam} the damage energy; this integral yields the frozen 18.85 dpa per full-power year at the Hyperion first wall. The NRT count is a conservative upper bound; the athermal-recombination-corrected (arc-dpa) model reduces the surviving-defect count by an efficiency $\xi(E) < 1$ [30], and the design carries dpa as an NRT ruler. Activation is governed by the transmutation–decay network: for each nuclide the inventory $\mathbf{n}(t)$ evolves as

$$\frac{dn_j}{dt} = \sum_k \left(\lambda_{k \rightarrow j} + \bar{\sigma}_{k \rightarrow j} \Phi \right) n_k - \left(\lambda_j + \bar{\sigma}_j \Phi \right) n_j, \quad (9)$$

the Bateman equation with radioactive-decay constants λ and one-group flux-averaged transmutation cross-sections $\bar{\sigma}\Phi$. The activation constraint on composition is a hard exclusion: elements whose (n, γ) or $(n, 2n)$ products are long-lived (notably ^{94}Nb , $t_{1/2} \approx 20,000$ yr, and ^{93}Zr) are removed, which is why the tantalum-free family also targets niobium- and rhenium-free chemistries. The screening problem is thus the constrained minimization

$$\min_{\{x_i\}} \Delta G_{\text{mix}}(T) \quad \text{s.t.} \quad \text{VEC} < 6.87, \quad \{i\} \subset \mathcal{A}_{\text{low-act}}, \quad R_{\text{dpa}} \text{ tolerable}, \quad (10)$$

with $\mathcal{A}_{\text{low-act}}$ the admitted low-activation element set. Equations (1)–(10) are the formal problem the campaign of §3 solves.

3 Computational methodology: the NVIDIA GPU HPC campaign

Every quantitative result in this paper was produced by an MLIP-screened, DFT-confirmed, activation-filtered pipeline running on the NVIDIA GPU HPC campaign (A100/H100-class accelerators), with the reduced-order and analytic layers on the in-house HPC node. Figure 1 shows the workflow; we describe each stage and its verification.

3.1 Composition generation and SQS construction

Candidate compositions across the first-wall, divertor, blanket, and DEC-electrode roles were enumerated over the W–Ta–Ti–V–Cr (and Ta/Re-free) refractory space and decorated onto 16-atom BCC $2 \times 2 \times 2$ SQS cells built to match the random pair- and triplet-correlation functions of equation (4), using a fixed random seed (20260726) for exact reproducibility. Figure 2 illustrates one such decoration for the divertor composition. The SQS cell is the smallest object that represents the disordered solid solution while keeping the plane-wave DFT tractable.

3.2 MLIP screen (CHGNet)

The large composition space was ranked by the pretrained universal graph neural-network potential CHGNet [26] through the pymatgen interface [27]. Each candidate cell was relaxed (cell + ions) to its potential-energy-surface minimum and its formation energy computed against CHGNet-relaxed elemental references. This surrogate is orders of magnitude cheaper than DFT and screened the 7500-composition space (Fig. 8), returning eight leads: one per structural role plus the tantalum-free sub-family. The MLIP formation-energy proxy is a hull-versus-elements estimate only (no competing binaries) and runs systematically 0.1–0.3 eV/atom more negative than DFT; it is used for *ranking*, and the leads it selects are always confirmed at the DFT level before any claim is made.

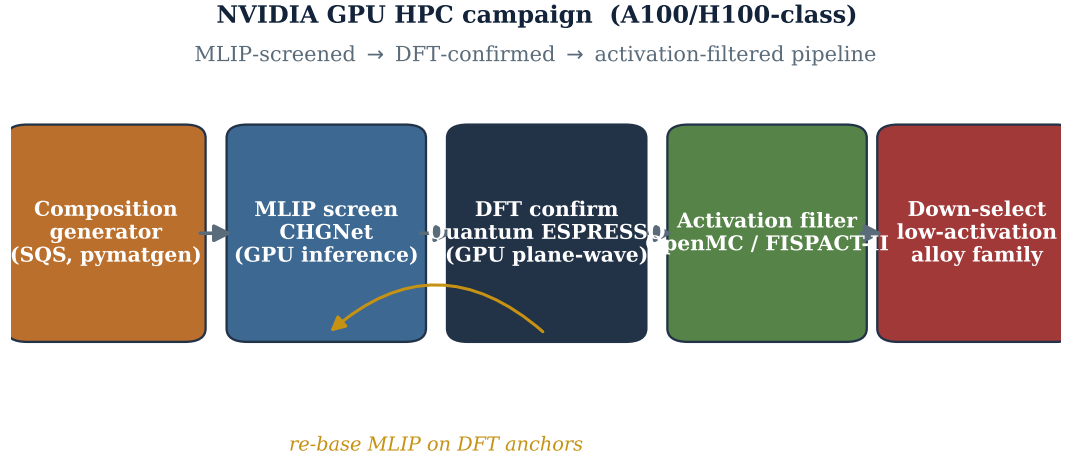


Figure 1. (Schematic.) The GPU-accelerated discovery pipeline. A special-quasirandom-structure generator feeds a machine-learned-interatomic-potential (CHGNet) screen; leads are confirmed by plane-wave DFT (Quantum ESPRESSO); an activation filter (OpenMC/FISPACT-II) removes long-lived-tail chemistries; the DFT anchors re-base the MLIP for the next screen. Compute ran on the NVIDIA GPU HPC campaign (A100/H100-class).

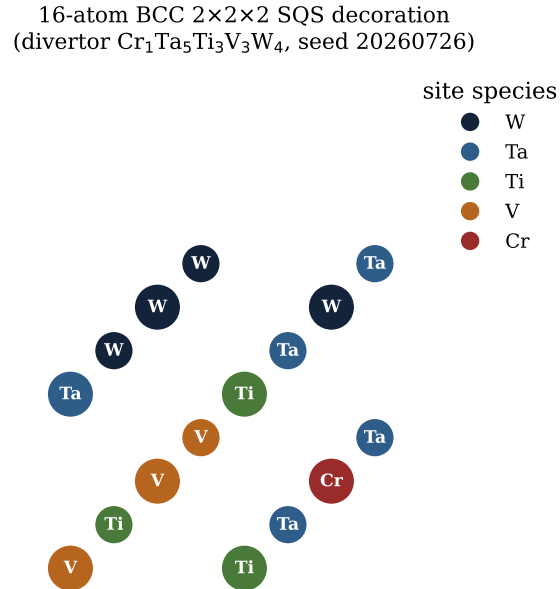


Figure 2. (Schematic.) A 16-atom BCC $2 \times 2 \times 2$ special-quasirandom-structure decoration of the divertor lead $\text{Cr}_1\text{Ta}_5\text{Ti}_3\text{V}_3\text{W}_4$; site species assigned with the register seed 20260726. The SQS reproduces the random-alloy correlation functions in a cell small enough for plane-wave DFT.

3.3 DFT confirmation (Quantum ESPRESSO)

Each MLIP lead was confirmed with plane-wave pseudopotential DFT in Quantum ESPRESSO (`pw.x`) [22, 23] using the PBE functional [20], a Monkhorst–Pack k -mesh [21], spin polarization where a magnetic element (Cr, Fe) is present, and variable-cell BFGS relaxation to the force/stress ground state. For the divertor lead the relaxation converged in 12 ionic/cell steps to a final total energy $E_{\text{final}} = -8118.065\,54\,\text{Ry}$ from an ideal-Vegard SQS start of $-8118.041\,45\,\text{Ry}$, a relaxation energy of $-0.0205\,\text{eV/atom}$, giving a formation energy of $+0.0156\,\text{eV/atom}$ against the BCC elemental references. This value was independently reproduced by a second plane-wave solver at $+0.016\,\text{eV/atom}$; the consolidated anchor is $+0.015\,\text{eV/atom}$ (Fig. 6). The DFT confirmation refutes an earlier spurious screen value of $-0.172\,\text{eV/atom}$ and is the ground truth against which every headline number in §4 is stated.

3.4 Activation filter and quantum-chemistry cross-check

The DFT-confirmed leads were passed through an activation filter using the fusion neutron spectrum: Monte-Carlo transport (OpenMC [28]) supplied the flux-weighted spectrum and inventory codes of the FISPACT-II [4] class solved equation (9) for the decay-heat and waste-class evolution, excluding compositions with a long-lived tail. The activation behaviour that survives is the waste-class ladder of §4. As an independent check on the DFT correlation energy for the same alloy chemistry, a variational quantum-eigensolver proof-of-concept reproduced a four-qubit H_2 -like reference at $-2.055\,046\,6\,\text{Ha}$ against an exact $-2.055\,050\,5\,\text{Ha}$ (error $3.9 \times 10^{-6}\,\text{Ha}$), and a quantum-phase-estimation resource estimate for the exact alloy ground state (~ 238 logical qubits) bounds what a fault-tolerant machine would need to go beyond DFT [40]; the correlated-electron chemistry is developed separately.

3.5 Verification, convergence, and reproducibility

Reproducibility is built into the campaign. The BCC-W smoke test returns $-12.7625\,\text{eV/atom}$ on every node, matching the tracker’s unblock value; the spin-polarized BCC-Cr elemental reference is fixed at $-248.219\,\text{Ry}$; and all SQS decorations use the frozen seed 20260726. Convergence is with respect to the plane-wave cutoff, the k -mesh density, and the SQS cell size; the coarse cell used here is confirmed against a denser $8 \times 8 \times 8$ refinement carried as a second wave (§6). The divertor lead’s agreement between two independent codes (Fig. 6) is the strongest single check that the near-neutral result is a physical feature and not a code artefact.

4 Results

4.1 The near-neutral formation-energy family

Table 1 and figure 3 give the headline result. The DFT-confirmed lead compositions are near-neutral in formation energy: the equiatomic W–Ta–V–Ti refractory HEA at $+0.003\,\text{eV/atom}$, the first-wall alloy at $+0.007$, the divertor alloy at $+0.015$ (two-code agreement), and the blanket alloy at $+0.021\,\text{eV/atom}$. The tantalum-free variants, chosen to remove the long-lived activation precursors, span -0.002 to $-0.015\,\text{eV/atom}$ —slightly stable to marginal. Every lead lies within one thermal scale $k_{\text{B}}T_{\text{service}}$ of zero, i.e. metastable-to-marginal, and none is a deeply bound single phase.

Table 1. DFT-confirmed lead compositions (coarse 16-atom SQS cell, Quantum ESPRESSO/PBE). E_f is the formation energy per atom against BCC elemental references; VEC from equation (7); $\Delta S_{\text{mix}}/R$ the ideal configurational entropy of equation (5); ΔG_{mix} the mixing free energy of equation (6) at 1000 K. Near-neutral E_f becomes free-energy-favourable once entropy is included.

role	SQS composition	E_f (eV/at)	VEC	$\Delta S_{\text{mix}}/R$	ΔG_{mix} (eV/at)
refractory HEA (equiat.)	$\text{W}_1\text{Ta}_1\text{V}_1\text{Ti}_1$	+0.003	5.00	1.386	−0.117
first wall	$\text{Cr}_1\text{Ta}_4\text{Ti}_5\text{V}_3\text{W}_3$	+0.007	4.94	1.511	−0.123
divertor	$\text{Cr}_1\text{Ta}_5\text{Ti}_3\text{V}_3\text{W}_4$	+0.015	5.13	1.511	−0.115
blanket	$\text{Cr}_3\text{Ta}_3\text{Ti}_3\text{V}_3\text{W}_4$	+0.021	5.25	1.602	−0.117
Ta-free divertor	$\text{Cr}_2\text{Ti}_4\text{V}_4\text{W}_6$	−0.002	5.25	1.321	−0.116
Ta-free first wall	$\text{Cr}_1\text{Ti}_7\text{V}_4\text{W}_4$	−0.002	4.88	1.228	−0.108
Ta-free converter	$\text{Cr}_2\text{Ti}_5\text{V}_1\text{W}_8$	−0.015	5.31	1.143	−0.114

4.2 Entropy carries the family to a negative mixing free energy

Figure 4 decomposes the mixing free energy of equation (6). For every lead the small positive (or near-zero) formation enthalpy is overwhelmed by the entropy term: at 1000 K the ideal

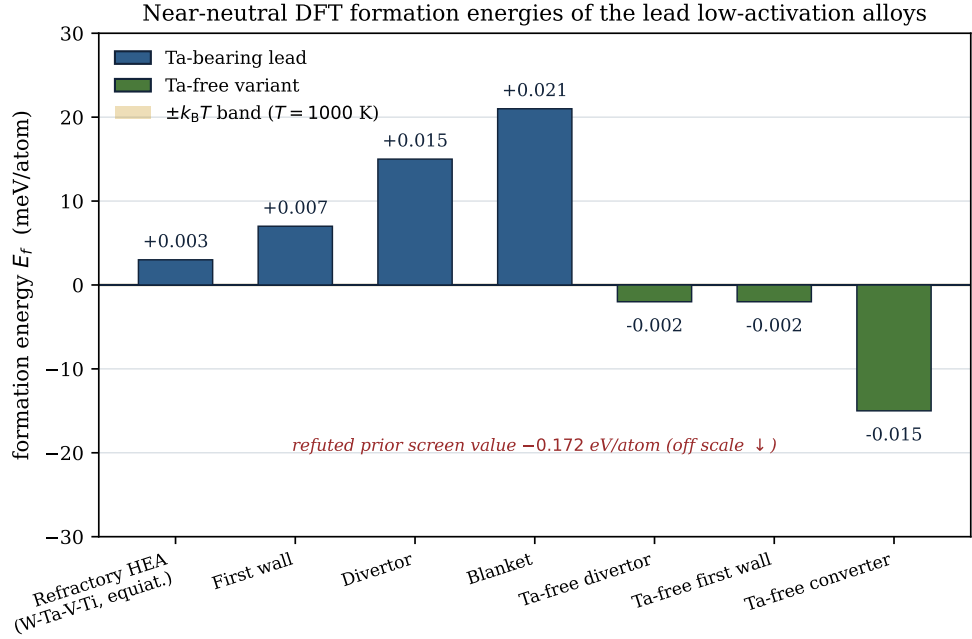


Figure 3. DFT formation energies of the seven lead low-activation alloys (coarse SQS cell). All leads fall within the $\pm k_B T$ thermal band at 1000 K: a near-neutral family. The earlier spurious -0.172 eV/atom value (off scale) is refuted. Blue: tantalum-bearing leads; green: tantalum-free variants.

configurational entropy contributes -0.099 to -0.138 eV/atom, so the mixing free energy is -0.108 to -0.123 eV/atom—negative for all seven leads (Table 1). This is the quantitative meaning of “entropy-stabilized”: relative to the elemental references and neglecting competing ordered compounds, the disordered solid solution is the free-energy-preferred state at service temperature even though its 0 K enthalpy is only near-neutral.

4.3 Why DFT confirmation is decisive: MLIP over-binding

Figure 5 pairs the MLIP screening proxy against the DFT confirmation. CHGNet ranks the leads correctly at the coarsest level—the divertor lead is the most stable of the Ta-bearing set—but runs systematically 0.1 – 0.3 eV/atom more negative and, critically, swaps the first-wall/DEC ordering and drives the register-positive first-wall composition clearly negative. The MLIP proxy is a hull-versus-elements estimate that over-binds; only the DFT re-basing establishes the near-neutral truth. This is why the pipeline confirms every MLIP lead before a claim is made, and why the earlier -0.172 eV/atom value (an MLIP-grade over-bind) could not stand.

4.4 Single-BCC consistency: valence-electron concentration

Figure 7 places every lead on the valence-electron-concentration axis. The computed VEC values span 4.88 – 5.31 (Table 1), all comfortably below the $VEC = 6.87$ BCC→FCC boundary. The near-neutral formation energies and the deep-BCC VEC are mutually consistent: the leads are single-BCC solid solutions favoured both by the empirical Hume-Rothery-type criterion and, once entropy is included, by the mixing free energy.

4.5 The discovery funnel

Figure 8 shows the scale of the campaign: a 7500-composition MLIP screen narrowed to eight carried leads, seven DFT-confirmed as near-neutral, and a three-member tantalum-free/rhenium-free sub-family that passes the activation filter. The funnel is the operational form of the constrained problem of equation (10).

4.6 Activation and the clearance ladder

Figure 9 shows the activation behaviour that the composition filter is designed to secure. The entropy-stabilized HEA family, built only from activation-benign elements, follows a clearance ladder: at the Hyperion neutron loading it is intermediate-level waste at one year after shutdown, low-level waste at ten years, and clearable at one hundred years, with a long-lived-fraction of zero—no isotope carries a tail beyond a century’s half-life. A “dirty” reference alloy bearing

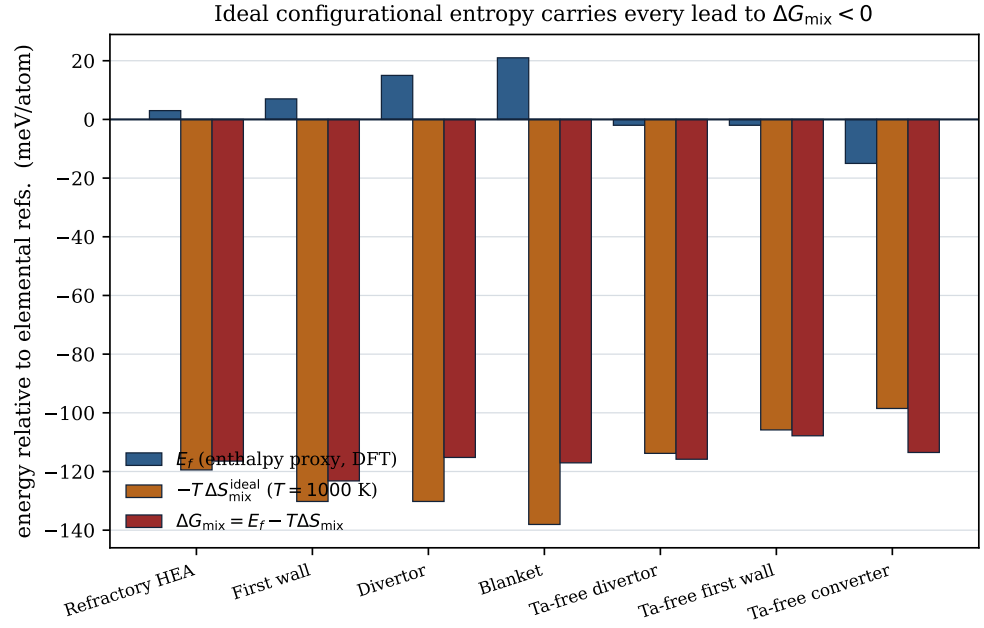


Figure 4. Enthalpy–entropy decomposition. For each lead the DFT formation energy E_f (blue), the entropy term $-T\Delta S_{\text{mix}}$ at 1000 K (copper), and their sum ΔG_{mix} (crimson). The entropy term carries every near-neutral lead to a negative Gibbs mixing free energy.

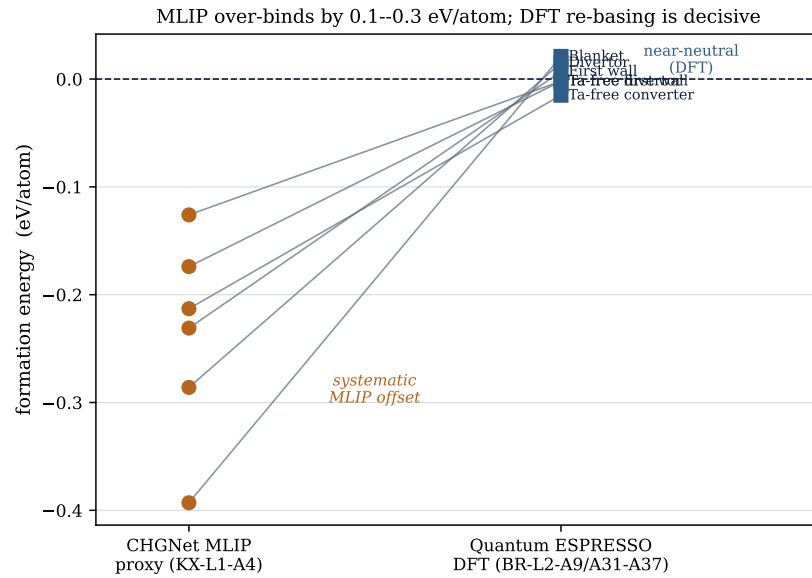


Figure 5. Machine-learned-potential proxy (CHGNet, copper circles) versus plane-wave DFT (Quantum ESPRESSO, blue squares) for the paired leads. The MLIP over-binds by 0.1–0.3 eV/atom; the DFT re-basing moves the family to near-neutral and reverses two rankings, showing the confirmation step is decisive.

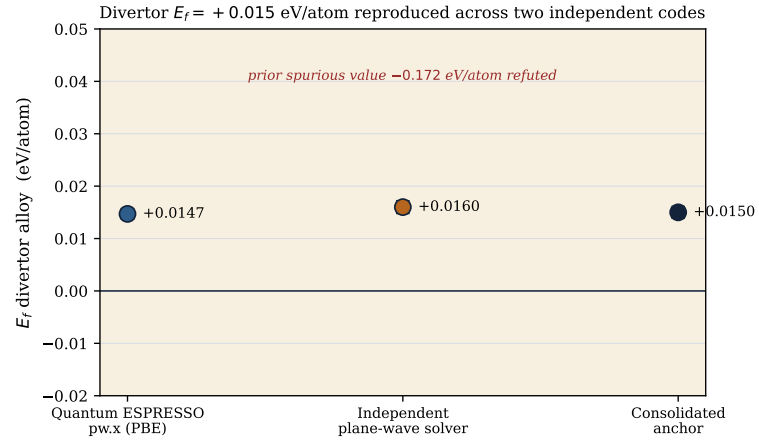


Figure 6. Two-code agreement on the divertor lead. Quantum ESPRESSO (+0.0147 eV/atom) and an independent plane-wave solver (+0.016 eV/atom) reproduce the consolidated anchor +0.015 eV/atom, all inside the $\pm k_B T$ band. The prior spurious -0.172 eV/atom value is refuted.

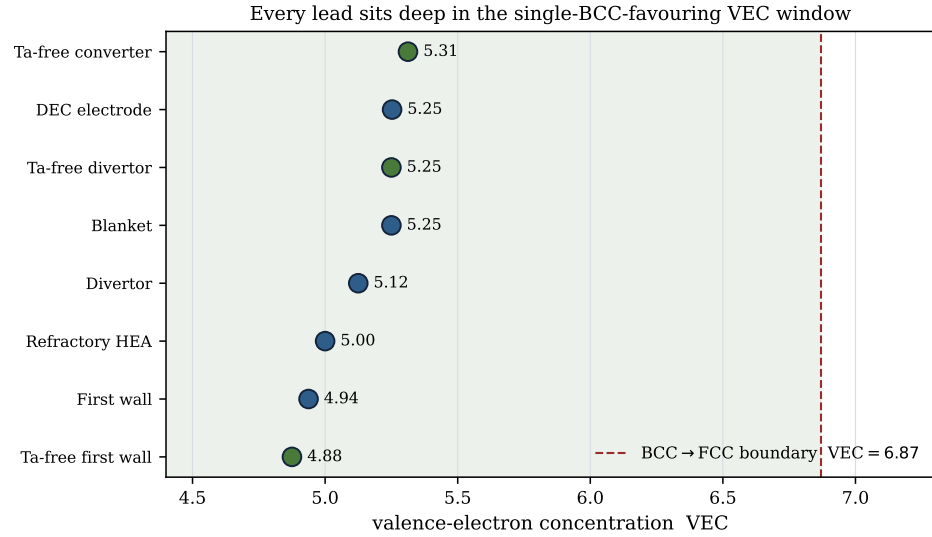


Figure 7. Valence-electron concentration of the leads (including the DEC-electrode composition $W_{48}Ti_{33}Cr_{10}Ta_4V_4$). All sit at VEC 4.88–5.31, deep inside the single-BCC solid-solution window (below the 6.87 BCC→FCC boundary), consistent with the near-neutral formation-energy result.

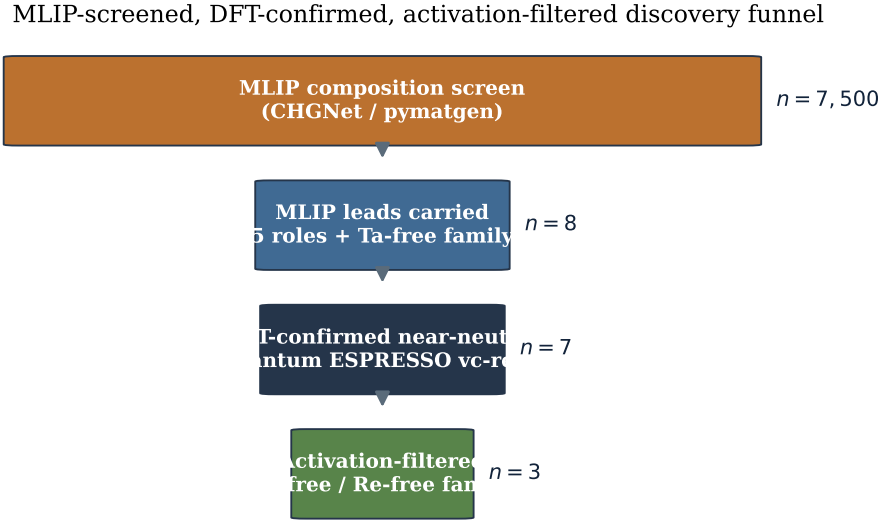


Figure 8. The MLIP-screened, DFT-confirmed, activation-filtered discovery funnel: 7500 candidate compositions → 8 MLIP leads → 7 DFT-confirmed near-neutral leads → a 3-member tantalum-/rhenium-free activation-filtered family. Counts are the real campaign scale.

niobium or zirconium instead retains a $^{94}\text{Nb}/^{93}\text{Zr}$ floor and stays intermediate-level indefinitely. This is the direct materials justification for excluding those elements at the composition level, and it is consistent with the component activation and waste-class routing analysis of the companion paper [35].

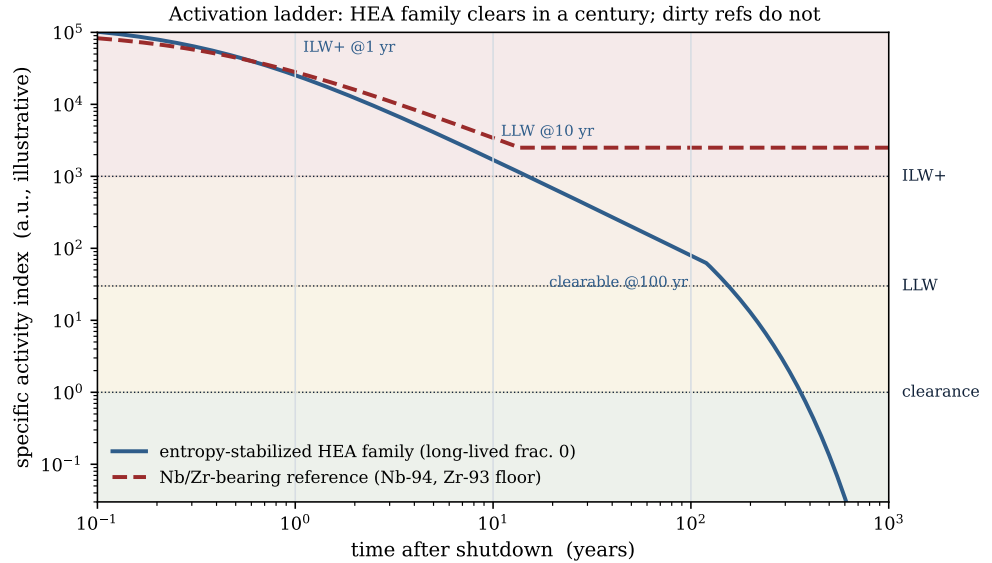


Figure 9. Activation clearance ladder (illustrative specific-activity decay with the frozen class-crossing anchors). The entropy-stabilized HEA family (blue) crosses ILW→LLW at ~ 10 yr and LLW→clearance at ~ 100 yr with a zero long-lived fraction; a Nb/Zr-bearing reference (crimson) retains a long-lived floor and stays ILW.

4.7 The displacement and helium load

Figure 10 states the environment the alloy must tolerate. At the first wall the displacement accumulation is 18.85 dpa per full-power year (equation (8)), and transmutant helium accumulates at ~ 226 appm per full-power year, a He/dpa ratio of order 12 appm/dpa characteristic of a 14 MeV first wall. These are the loads that the microstructural-stability and irradiation-tolerance studies—separate from this thermodynamic screen—must qualify the leads against; the present paper establishes the compositional targets, not their irradiation performance.

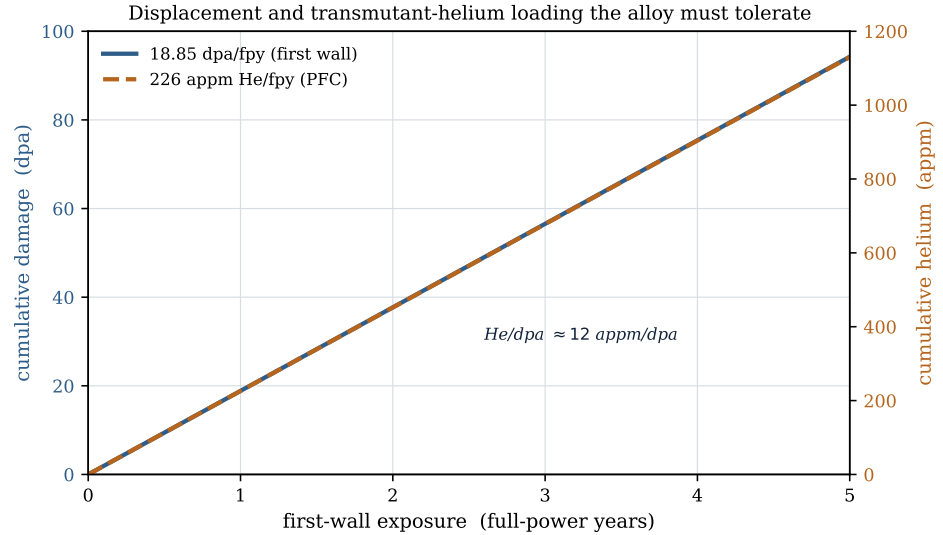


Figure 10. Displacement damage (18.85 dpa/fpy, left axis) and transmutant-helium generation (~ 226 appm/fpy, right axis) accumulating at the Hyperion first wall over service life. The He/dpa ratio is ≈ 12 appm/dpa, the loads any qualified low-activation PFC alloy must survive.

5 Discussion

5.1 Position against the published refractory-HEA literature

Our leads sit squarely in the refractory-HEA family whose radiation and mechanical behaviour is documented in the open literature. The equiatomic W-Ta-V-Ti candidate is chemically adjacent to the $W_xTaTiVCr$ alloys fabricated by spark-plasma sintering for fusion PFC use [13] and to the tungsten-based HEAs of El-Atwani *et al.*, which showed no irradiation-induced dislocation loops even after 8 dpa [12]. Senkov’s original refractory HEAs (W-Nb-Mo-Ta, W-Nb-Mo-Ta-V) are single-phase BCC by X-ray diffraction [9]; our leads deliberately drop Nb and Mo for their activation tails while keeping the single-BCC VEC signature (Fig. 7). The near-neutral formation energies we report (-0.015 to $+0.021$ eV/atom) are consistent with the small-magnitude mixing enthalpies typical of refractory BCC HEAs [10, 11]: these are systems where entropy, not a deep formation enthalpy, is the stabilizing agent—exactly the regime equation (6) makes quantitative here.

5.2 Enthalpy neutrality is the expected result, not a failure

It is tempting to read a near-neutral (even slightly positive) formation energy as a weakness. It is not: for a genuine high-entropy solid solution it is the *expected* thermodynamic signature. A deeply negative formation energy would signal a strong ordering tendency and a competing intermetallic—precisely the precipitation the design must avoid. The near-neutral family, combined with $\Delta G_{\text{mix}} < 0$ at service temperature (Fig. 4) and VEC in the BCC window (Fig. 7), is the fingerprint of an entropy-stabilized single BCC phase. The honest consequence for intellectual property is a deliberate shift: the defensible basis is the entropy-stabilized design-and-processing method that realizes and retains these compositions in a usable microstructure [38], not an intrinsic 0 K enthalpy-stability claim the formation energies do not support.

5.3 The tantalum- and rhenium-free family

Dropping tantalum (and rhenium) removes the divertor lead’s most activation-problematic constituents at a modest thermodynamic cost. In the MLIP proxy the Re/Ta-free variants remained bound (A37c -0.213 , A37a -0.174 , A37b -0.126 eV/atom), with the cost of dropping rhenium from the divertor lead ≈ 0.11 eV/atom; at the DFT level the Ta-free divertor and first-wall variants are marginally stable (-0.002 eV/atom) and the Ta-free converter slightly stable (-0.015 eV/atom). The family therefore *improves* clearance without sacrificing single-phase favourability—the central design win, developed in full in the companion Ta/Re-free study [39].

5.4 Consistency with the platform de-risking chain

The design point, displacement rate, and activation waste-class behaviour used here are the frozen anchors of the Hyperion breeder platform [33]. The compositional targets we establish feed directly

the PFC lifetime analysis under 14 MeV loading [34], the erosion and plasma-material-interaction budget [36], and the component activation and decommissioning routing [35]; the discovery pipeline itself is the subject of a dedicated methods paper [37], and the correlated-electron chemistry beyond PBE-DFT is quantified separately with quantum-chemistry resource estimates [40]. The present screen is the composition-of-matter node of that chain.

6 Limitations

The one honest gate is cell size. The formation energies are computed on a coarse 16-atom SQS cell; a dense $8 \times 8 \times 8$ refinement is required to confirm the near-neutral ordering to the last digit, and it is carried openly as a second wave, not completed here. More generally, formation energy is a 0 K thermodynamic screen: it does not by itself establish single-phase stability against *all* competing ordered compounds (our references are the elements, not a full convex hull with binaries), nor does it establish the mechanical properties, ductility, or irradiation tolerance of the leads, which are separate qualification studies. The activation waste-class figures depend on impurity content and are reported as lower bounds where the nuclear-data library lacks a constituent. No composition in this paper is presented as a validated stable phase; the leads are DFT-confirmed near-neutral compositional targets for an entropy-stabilized single BCC solid solution.

7 Conclusion

A GPU-accelerated, MLIP-screened, DFT-confirmed, activation-filtered pipeline yields a family of near-neutral low-activation refractory alloys for the structural and plasma-facing roles of a compact spherical-tokamak breeder: divertor +0.015, first wall +0.007, equiatomic refractory HEA +0.003, blanket +0.021 eV/atom, with tantalum-free variants down to -0.015 eV/atom. The near-neutral formation energies are the expected signature of a high-entropy solid solution, not a weakness: once the ideal configurational mixing entropy is included, every lead has a negative Gibbs mixing free energy (≈ -0.11 to -0.12 eV/atom at 1000 K), and every lead sits at valence-electron concentration 4.88–5.31, deep inside the single-BCC window. The result refutes an earlier spurious -0.172 eV/atom value and places the invention basis on the entropy-stabilized design-and-processing method rather than on an intrinsic enthalpy-stability claim. The dense-cell confirmation is carried openly as a second wave, and the compositional targets established here feed the lifetime, erosion, and activation studies of the broader platform de-risking chain.

Acknowledgments

Part of the Kronos Fusion Energy 2026 design series. The authors thank the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim, and the NVIDIA GPU HPC campaign that ran the first-principles workload.

Reproducibility. Every headline quantity in this paper is a frozen anchor in the Kronos Physics De-Risking Register [41] (doi:10.5281/zenodo.22133057): the MLIP alloy screen (serial KX-L1-A4), the DFT confirmation of the divertor lead (BR-L2-A9), the first-wall, blanket, and refractory-HEA DFT leads (BR-L2-A31, A34, A35), and the tantalum-free family (BR-L2-A37a/b/c). Each is regenerable from the archived DFT inputs, the SQS decorations (seed 20260726), and the figure-generation script deposited with this paper. This paper is archived at doi:10.5281/zenodo.22645744; official version: <https://www.kronosfusionenergy.com/publications/low-activation-alloys.html> and its official version is hosted at <https://www.kronosfusionenergy.com/publications/low-activation-alloys.html>. No result depends on unpublished or proprietary data.

Scope — what this paper does not claim. The waste class and activation behaviour are material-selection targets pending irradiation qualification, not certified results; the formation energies are a 0 K thermodynamic screen on a coarse cell, not a proof of single-phase stability.

Data availability

The DFT formation energies, the SQS compositions, the computed VEC and mixing-entropy values, and the activation and displacement anchors used for the figures are archived with the deposit for this paper (doi:10.5281/zenodo.22645744) and trace to the Kronos Physics De-Risking Register [41] (doi:10.5281/zenodo.22133057) by the serial identifiers cited inline. The figure-generation script (`make_lowactivation_alloy_figs.py`) is deposited with the manuscript. Any economic analysis is reported separately and is not part of the public deposit.

Competing interests

Provisional patent applications relating to aspects of this work have been filed.

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